

SCQ (Single Correct Type) :

- Sum the series : $\tan^{-1}\left(\frac{4}{1+3 \cdot 4}\right) + \tan^{-1}\left(\frac{6}{1+8 \cdot 9}\right) + \tan^{-1}\left(\frac{8}{1+15 \cdot 16}\right) + \dots \infty$ is :
 (A) $\cot^{-1}(2)$ (B) $\tan^{-1}(2)$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$
- The number of ordered pair(s) (x, y) which satisfy $y = \tan^{-1} \tan x$ and $16(x^2 + y^2) - 48\pi x + 16\pi y + 31\pi^2 = 0$, is :
 (A) 0 (B) 1 (C) 2 (D) 3
- Number of solution(s) of the equation $\cos^{-1}(1-x) - 2\cos^{-1}x = \frac{\pi}{2}$ is :
 (A) 3 (B) 2 (C) 1 (D) 0
- Let triangle ABC be an isosceles triangle with $AB = AC$. Suppose that the angle bisector of its angle B meets the side AC at a point D such that $BC = BD + AD$. Measure of the angle A in degrees, is :
 (A) 80 (B) 100 (C) 110 (D) 130
- If $\tan\left(\frac{\pi}{12} - x\right)$, $\tan\frac{\pi}{12}$, $\tan\left(\frac{\pi}{12} + x\right)$ in order are three consecutive terms of a G.P. then sum of all the solutions in $[0, 314]$ is $k\pi$. The value of k is :
 (A) 4950 (B) 5050 (C) 2525 (D) 5010
- If $a_i, i = 1, 2, 3, 4$ be four real members of the same sign, then the minimum value of $\sum \frac{a_i}{a_j}, i, j \in \{1, 2, 3, 4\}, i \neq j$ is:
 (A) 6 (B) 8 (C) 12 (D) 24
- Let T_r be the r^{th} term of an A.P. whose first term is $-\frac{1}{2}$ and common difference is 1, then $\sum_{r=1}^n \sqrt{1 + T_r T_{r+1} T_{r+2} T_{r+3}} =$
 (A) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4}$ (B) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4} + \frac{1}{4}$
 (C) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4} + \frac{1}{2}$ (D) $\frac{n(n+1)(2n+1)}{12} - \frac{5n}{8} + 1$

8. Let $A = [a_{ij}]_{3 \times 3}$ be such that $a_{ij} = \begin{cases} 3; & \text{when } \hat{i} = \hat{j} \\ 0; & \hat{i} \neq \hat{j} \end{cases}$, then $\left\{ \frac{\det(\text{adj}(\text{adj} A))}{5} \right\}$ equals :

(where $\{\cdot\}$ denotes fractional part function)

- (A) $\frac{2}{5}$ (B) $\frac{1}{5}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

9. Let matrix $A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ where $x, y, z \in \mathbb{N}$. If $\det. (\text{adj.} (\text{adj.} A)) = 2^8 \cdot 3^4$ then the number of such

matrices A is :

[Note : adj. A denotes adjoint of square matrix A.]

- (A) 220 (B) 45 (C) 55 (D) 110

10. If $\text{Trace}(A) > 0$ and $abc = 1$ where $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ and A is orthogonal matrix, then find the value of a^2

$+ b^2 + c^2$:

- (A) 1 (B) 2 (C) 3 (D) 0

11. If a variable circle S touches $S_1 : |z - z_1| = r_1$ internally and $S_2 : |z - z_2| = r_2$ externally while the curves S_1 & S_2 touch internally to each other. Then the eccentricity of the locus of the centre of the curve S is equal to

- (A) $\frac{r_1}{r_2}$ (B) $\frac{r_2}{r_1}$ (C) $\frac{r_1 + r_2}{r_1 - r_2}$ (D) $\frac{r_1 - r_2}{r_1 + r_2}$

12. Let $N = 2^{1224} - 1$, $\alpha = 2^{153} + 2^{77} + 1$ and $\beta = 2^{408} - 2^{204} + 1$. Then which of the following statement is correct ?

- (A) α divides N but β does not (B) β divides N but α does not
(C) α and β both divide N (D) Neither α nor β divides N

MCQ (One or more than one correct) :

13. Which of the following alternative (s) is/are incorrect ?

- (A) Number of values of x satisfying simultaneously $\sin^{-1}x = 2 \tan^{-1}x$ and

$$\tan^{-1} \sqrt{x(x-1)} + \text{cosec}^{-1} \sqrt{1+x-x^2} = \frac{\pi}{2}, \text{ is } 2$$

- (B) Number of values of x satisfying simultaneously

$$\sin^{-1}x = 2 \tan^{-1}x \text{ and } \tan^{-1} \tan^{-1} \sqrt{x(x-1)} + \text{cosec}^{-1} \sqrt{1+x-x^2} = \frac{\pi}{2}, \text{ is greater than } 2$$

- (C) The complete solution set of the equation $\sin^{-1} \sqrt{1-x^2} + \cos^{-1} x = \cot^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) - \sin^{-1} x$ is $[-1, 0]$

(D) The complete solution set of the equation $\sin^{-1} \sqrt{1-x^2} + \cos^{-1} x = \cot^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) - \sin^{-1} x$ is $[-1, 1]$

14. In a triangle ABC, If D is mid point of side BC and AD is perpendicular to AC, then the value of $\cos A \cdot \cos C$ is

(A) $\frac{2b^2}{ac}$ (B) $\frac{2(a^2 - c^2)}{3bc}$ (C) $-\frac{2b^2}{ac}$ (D) $\frac{2(c^2 - a^2)}{3ac}$

Numerical Based Questions :

15. Find number of solutions of the equation

$$\sin^{-1}(|\log_6^2(\cos x) - 1|) + \cos^{-1}(|3\log_6^2(\cos x) - 7|) = \frac{\pi}{2}, \text{ if } x \in [0, 4\pi].$$

16. Range of function $f(x) = |6\sin^{-1}x - \pi| + |6\cos^{-1}x - \pi|$ is $[a\pi, b\pi]$, find the value of $(a + b)$

17. If the quadratic equation $ax^2 + bx + c = 0$ has equal roots where a, b, c denotes the lengths of the sides opposite to vertex A, B and C of the $\triangle ABC$ respectively. Find the number of integers in the range of $\frac{\sin A}{\sin C} + \frac{\sin C}{\sin A}$.

18. If $z_1, z_2, z_3 \in \mathbb{C}$ satisfying $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1 z_2 z_3 = 1$. Also $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$. Then find the value of $[|z_1 + z_2^2 + z_3^3|]$, where $[.]$ denotes the greatest integer function.

19. Consider curves $C_1: |\arg(z - 2)| = \frac{\pi}{4}$; $C_2: |\arg(z + 2)| = \frac{3\pi}{4}$ and $C_3: |z - \alpha i| = 4\sqrt{2}$, $\alpha \in \mathbb{R}$ on the complex plane such that C_3 touches both C_1 and C_2 , then find the sum of absolute value(s) of α

20. Let $\alpha_n = (2 + \sqrt{3})^n$. If $[.]$ denotes greatest integer function and $n \in \mathbb{N}$ then find the value of $\lim_{n \rightarrow \infty} (\alpha_n - [\alpha_n])$ is equal to

ANSWER KEY OF CAPS-4+

1.	(A)	2.	(D)	3.	(C)	4.	(B)	5.	(A)	6.	(C)	7.	(C)
8.	(B)	9.	(C)	10.	(A)	11.	(D)	12.	(C)	13.	(BCD)	14.	(CD)
15.	4	16.	10	17.	3	18.	2	19.	12	20.	(1)		